

EDS Assignment No. 2

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CS2-C24

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Topic : Enron Email Datasheet

Problem Statement :

To analyze the given SMS dataset using Pandas and NumPy by performing data exploration, statistical analysis, and text-based operations such as classification of messages (spam/ham), computation of message length statistics, pattern detection, and transformation of categorical data into numerical form.

20 Problem Statements :

1. Determine the total number of message present in the dataset :

```
import pandas as pd
import numpy as np

from google.colab import files
uploaded = files.upload()

df = pd.read_csv('mail_data.csv')
df.head()
```

Choose files mail_data.csv

mail_data.csv(text/csv) - 485702 bytes, last modified: 30/04/2026 - 100% done
Saving mail_data.csv to mail_data (1).csv

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
print(df.shape[0])
```

5572

2. Count the number of messages belonging to each category (spam and ham) :

```
print(df['Category'].value_counts())
```

```
Category
ham      4825
spam     747
Name: count, dtype: int64
```

3. Calculate the percentage distribution of spam and ham messages :

```
print(df['Category'].value_counts(normalize=True) * 100)
```

```
Category
ham      86.593683
spam     13.406317
Name: proportion, dtype: float64
```

4. Create a new feature representing the length of each message :

```
df['length'] = df['Message'].apply(len)
print(df.head())
```

	Category	Message	length
0	ham	Go until jurong point, crazy.. Available only ...	111
1	ham	Ok lar... Joking wif u oni...	29
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	155
3	ham	U dun say so early hor... U c already then say...	49
4	ham	Nah I don't think he goes to usf, he lives aro...	61

5. Find the maximum length of messages in the dataset :

```
df['length'] = df['Message'].apply(len)
print(df['length'].max())
```

910

6. Find the minimum length of messages in the dataset :

```
df['length'] = df['Message'].apply(len)
print(df['length'].min())
```

2

7. Compute the average (mean) length of all messages :

```
 df['length'] = df['Message'].apply(len)
print(df['length'].mean())
```

... 80.36898779612348

8. Determine the median length of messages using numpy :

```
df['length'] = df['Message'].apply(len)
print(np.median(df['length']))
```

... 61.0

9. Calculate the Standard Deviation of message lengths :

```
df['length'] = df['Message'].apply(len)
print(np.std(df['length']))
```

... 59.921567785815725

10. Count how many messages contain the word “free” (case insensitive) :

```
print(df['Message'].str.contains('free', case=False).sum())
```

... 265

11. Extract all messages that are classified as spam :

```

▶ spam = df[df['Category'] == 'spam']
print(spam.head())

```

```

...   Category      Message  length
2    spam  Free entry in 2 a wkly comp to win FA Cup fina...    155
5    spam  FreeMsg Hey there darling it's been 3 week's n...    147
8    spam  WINNER!! As a valued network customer you have...    157
9    spam  Had your mobile 11 months or more? U R entitle...    154
11   spam  SIX chances to win CASH! From 100 to 20,000 po...    136

```

12. Extract all messages that are classified as ham:

```

▶ ham = df[df['Category'] == 'ham']
print(ham.head())

```

```

...   Category      Message  length
0    ham  Go until jurong point, crazy.. Available only ...    111
1    ham                      Ok lar... Joking wif u oni...     29
3    ham  U dun say so early hor... U c already then say...     49
4    ham  Nah I don't think he goes to usf, he lives aro...     61
6    ham  Even my brother is not like to speak with me. ...     77

```

13. Compute the average message length for each category (spam Vs ham) :

```

▶ df['length'] = df['Message'].apply(len)
print(df.groupby('Category')['length'].mean())

```

```

...   Category
ham      71.44829
spam     137.98929
Name: length, dtype: float64

```

14. Identify the message with the maximum length:

```
df['length'] = df['Message'].apply(len)
print(df.loc[df['length'].idxmax()])
```

```
... Category      ham
   Message      For me the love should start with attraction.i...
   length      910
   Name: 1085, dtype: object
```

15. Identify the message with the minimum length :

```
df['length'] = df['Message'].apply(len)
print(df.loc[df['length'].idxmin()])
```

```
... Category      ham
   Message      Ok
   length      2
   Name: 1925, dtype: object
```

16. Count the number of messages that contain numerical digits :

```
print(df['Message'].str.contains(r'\d').sum())
```

```
... 1460
```

17. Count the number of messages that contain special characters :

```
print(df['Message'].str.contains(r'[@#$%^&*(),.?"':{}|<>]').sum())
```

```
... 4968
```

18. Convert the categorical labels (spam/ham) into numerical format using NumPy :

```
df['Category_num'] = np.where(df['Category'] == 'spam', 1, 0)
print(df.head())
```

```
...   Category      Message  length \
0      ham  Go until jurong point, crazy.. Available only ...    111
1      ham                Ok lar... Joking wif u oni...     29
2     spam  Free entry in 2 a wkly comp to win FA Cup fina...   155
3      ham  U dun say so early hor... U c already then say...    49
4      ham  Nah I don't think he goes to usf, he lives aro...    61

   Category_num
0              0
1              0
2              1
3              0
4              0
```

19. Find the correlation between message length and category :

```
df['length'] = df['Message'].apply(len)
df['Category_num'] = np.where(df['Category'] == 'spam', 1, 0)
print(df[['length', 'Category_num']].corr())
```

```
...      length  Category_num
length      1.000000      0.378359
Category_num 0.378359      1.000000
```

20. Retrieve the top 5 longest messages from the dataset :


```
df['length'] = df['Message'].apply(len)
print(df.nlargest(5, 'length'))
```

```
...      Category      Message  length \
1085      ham  For me the love should start with attraction.i...    910
1863      ham  The last thing i ever wanted to do was hurt yo...    790
2434      ham  Indians r poor but India is not a poor country...    629
1579      ham  How to Make a girl Happy? It's not at all diff...    611
2158      ham  Sad story of a Man - Last week was my b'day. M...    588

      Category_num
1085              0
1863              0
2434              0
1579              0
2158              0
```